

Calibrated decisions depend on the sampled specification

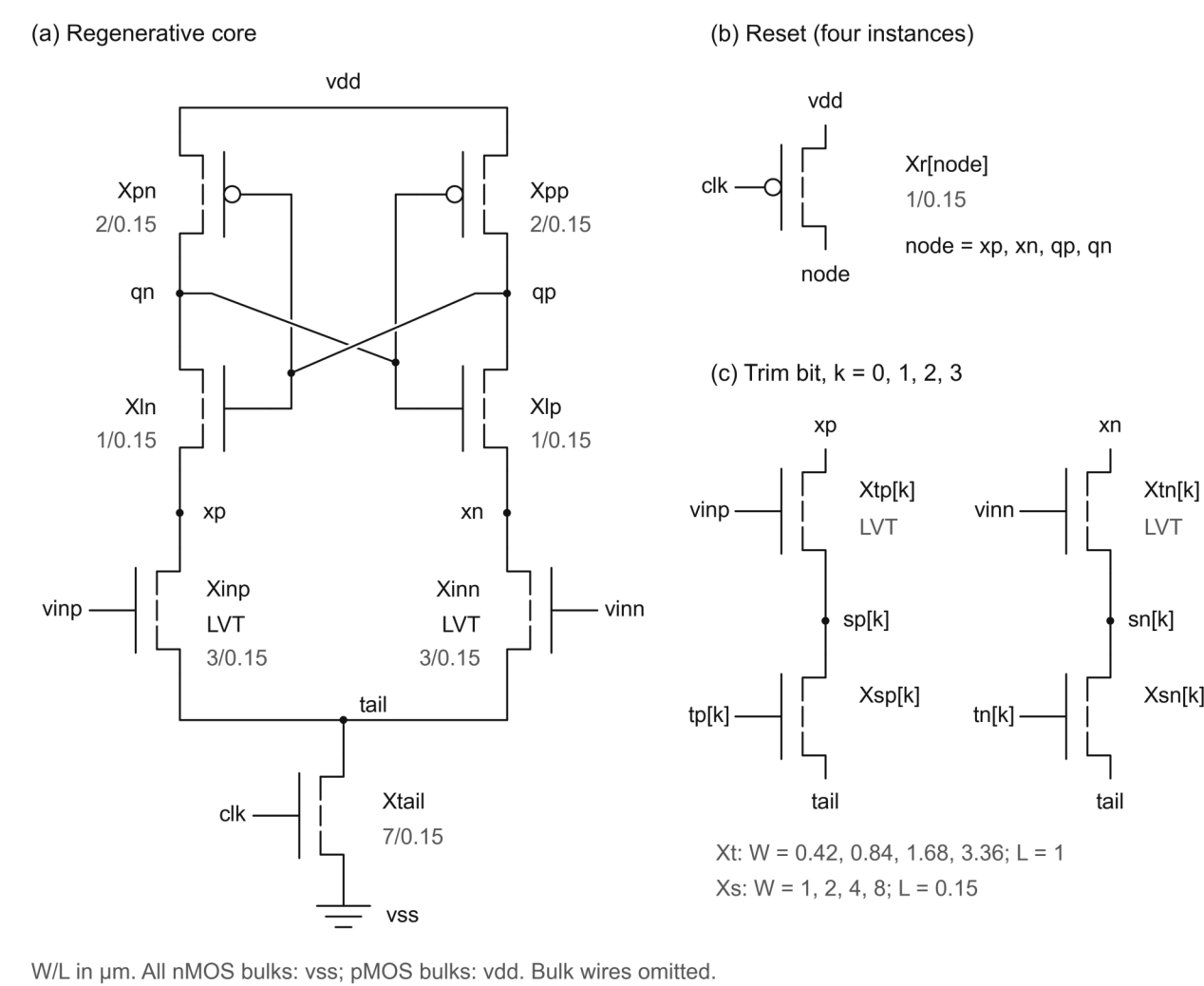
Comparator Atlas | SKY130 StrongARM characterization

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Circuit and decision contract

27-MOS StrongARM; LVT inputs, SVT latch.
Source-gated trim: sign + four magnitude bits.



Recorded circuit topology. Nine declared sizing candidates; the lower-energy control was evaluated after selection.

Correct, wrong or unresolved

A correct decision reaches complementary 80% / 20% output rails with the expected polarity before the deadline. Wrong polarity and unresolved outputs are counted separately.

Calibration and full-cycle core energy are evaluated from archived measurements. A later readout does not rerun SPICE or shorten the measured energy cycle.

Three sampled specifications

Calibrated SCHEMATIC: local_boundary; all 49 controlled-width-stress conditions. Both signs at finite sampled inputs through 30 mV; 1 ns deadline.

≥ 1 mV **NONE**

No feasible compared design

Mean energy / limits: not defined

≥ 3 mV **Selected: lvt_balanced_4b**

294/294 correct at all included samples

249.655 fJ mean core / cycle

≥ 30 mV **Control: lvt_base_3b**

98/98 correct at all included samples

150.531 fJ mean core / cycle

Post-hoc description of three compared designs: all-point qualification, then least mean energy. No interpolation or changed training selection.

Selected ≥ 3 mV band / same 294 samples

154.517 ps

363.217 fJ

Min. deadline - recorded decision time (sampled)

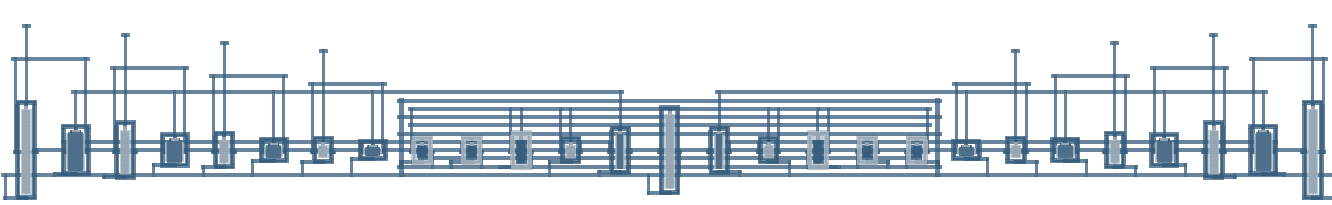
Maximum sampled core energy

20 wrong samples remain at 2 ns

Exactly the same keyed set at 1 ns and 2 ns, all at signed +/-1 mV. Locations and trim codes do not identify a physical failure cause.

Nominal layout and archived RC

Separate nominal, code-zero study: 45 PVT conditions x four signed inputs (-10, -3, +3, +10 mV). 180 points per mode; matched ngspice-47 settings.

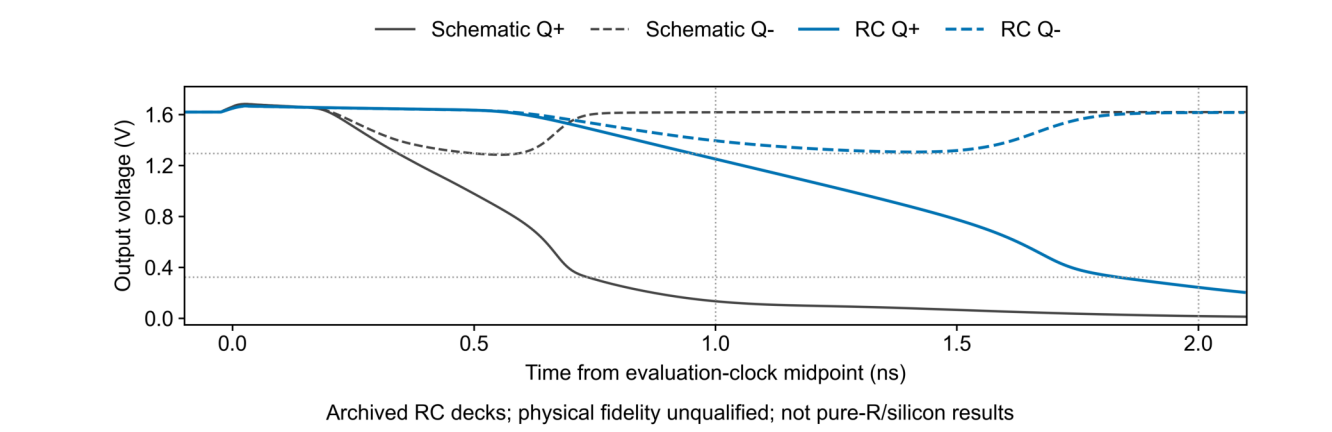


Original GDS geometry: 2207.088 μm^2 . DRC 0, LVS and negative controls: structural checks only.

Correct / 180	1 ns	2 ns
Schematic	180/180	180/180
Archived RC	156/180	180/180

Mean core: 245.84 fJ SC / 425.49 fJ RC

24 unresolved at 1 ns; zero wrong. The 2 ns deadline was separately, prospectively declared.



Worst sampled RC decision: 1.835 ns.
FS / 1.62 V / -40 C / -3 mV; 5 fF / output.

Archived RC-deck outcomes; model physical fidelity not yet qualified.

[Notebook](#) [Colab](#)

No qualified PEX or silicon prediction. C-only is not independent ground truth; RC-versus-C differences do not isolate resistance.

Finite samples, not foundry Monte Carlo/yield, noise/jitter/PVT confidence, timing signoff or a worst-cycle guarantee. Core excludes drivers and calibration/controller infrastructure.

References: Razavi (2015), DOI 10.1109/MSSC.2015.2418155; Li, Xu & Iizuka (2022), DOI 10.1007/s10470-022-01992-6.

Original code: MIT; SKY130: Apache-2.0. GitHub Copilot assisted implementation, experiment automation, figures and documentation; the author is responsible.